

Ultron — A Fine-Tuned Tool-Calling Agent for Authorized Penetration Testing

Ultron pairs a QLoRA fine-tuned **Qwen3.5-2B** model with a scoped, human-gated tool-calling orchestration layer to help authorized pentesters run recon, look up vulnerability references, and generate reports through natural-language instructions.

Project Documents

Document	Purpose
Ultron_Qwen3.5-2B_Finetuning_Report.docx	Model selection benchmarks + fine-tuning training results
MODEL_CARD.md	Model summary, intended use, limitations
DATASET_CARD.md	Fine-tuning dataset composition and provenance
TOOL_SCHEMA_REFERENCE.md	Every tool the model can call, with JSON schemas
API_INTEGRATION_GUIDE.md	How to serve the GGUF model and wire it to the orchestration layer
RESPONSIBLE_USE_POLICY.md	Authorization, human-in-the-loop, and logging guardrails
EVALUATION_REPORT.md	Current metrics + the plan to close the "no held-out eval" gap
EXECUTIVE_SUMMARY.docx	One-page overview for judges
Ultron_Slide_Deck.pptx	Presentation deck
DEMO_SCRIPT.md	Step-by-step walkthrough for the live demo

Quick Start

```
# 1. Serve the fine-tuned, quantized model
./llama-server -m ultron-qwen3.5-2b-toolcall.Q6_K.gguf --port 8080 --jinja

# 2. Point the orchestration layer at the server
export ULTRON_MODEL_ENDPOINT=http://localhost:8080/v1/chat/completions

# 3. Register an authorized engagement before running any tool
# (see TOOL_SCHEMA_REFERENCE.md → start_engagement)
```

See `API_INTEGRATION_GUIDE.md` for full serving and integration details.

Model Summary

- **Base:** unsloth/Qwen3.5-2B (Apache 2.0)
- **Fine-tuning:** QLoRA, 30 steps, 40,580 tokens, final loss 1.2061
- **Export:** GGUF, Q6_K quantization
- **Chosen for:** native function calling, best reasoning-per-parameter at the 2B tier, 262K context, Apache 2.0 license — see the full comparison in the fine-tuning report.

Status / Known Gaps

- No held-out evaluation set has been scored yet (`EVALUATION_REPORT.md`).
- Confirmation-gate UI for high-risk actions: `[status]` .
- See `RESPONSIBLE_USE_POLICY.md` §8 for the full open-items list.

Team

[Names / roles]

License

[Project license] — base model is Apache 2.0; see `MODEL_CARD.md` and `DATASET_CARD.md` for component-level licensing.

